Energy monitoring for a 3D printer park, built in Microsoft Fabric

Portfolio project by Vincent Antoine · github.com/v-nc/fabric-am-energy · October 2026

> **Personal portfolio project on simulated data.** It rebuilds, on Microsoft Fabric, an energy-monitoring system I built in 2019 for an internal park of 3D printers (OPC UA aggregation, InfluxDB, Grafana, alerts on over-long heat-ups). No real machine, customer or company data is used. DP-700 (Fabric Data Engineer Associate) is in preparation.

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OPC UA (10 simulated 3D printers + meters) → edge gateway → Eventstream → Lakehouse bronze → silver → gold → Direct Lake model → Power BI
Eventstream → Eventhouse (KQL) → Real-Time Dashboard and Activator e-mail alerts
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10,522 / 10,522
injected duplicates removed, none lost

100 of 108
heat-up overruns found, 0 false alarms

97.7 %
data completeness, by 3D printer and day

67 tests
local Spark and TypeScript, against a ground truth

WHAT I BUILT

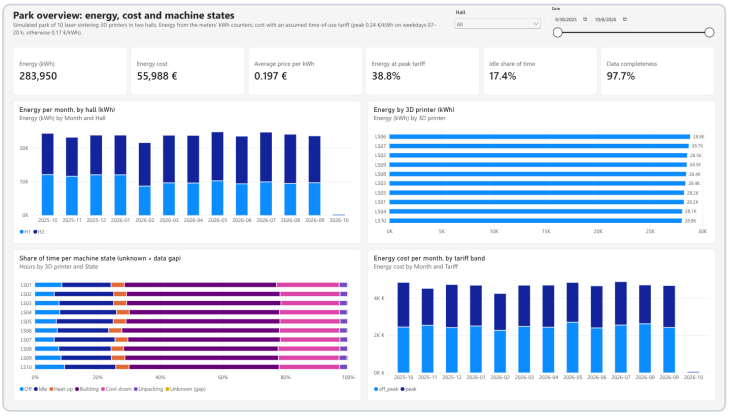
- **Edge:** OPC UA simulator and a TypeScript gateway with a SQLite store-and-forward buffer, per-meter sequence numbers and simulated network outages (late, out-of-order, duplicate events).
- **Ingestion:** Eventstream custom endpoint over the Event Hubs protocol; 12 months of history (5.2M rows, two schema versions) loaded in full and incrementally with a watermark.
- **Silver:** Spark Structured Streaming (`availableNow`), quality rules with a quarantine table, idempotent insert-only MERGE.
- **Gold:** star schema with energy from kWh counters, a time-of-use tariff in local time, build jobs, heat-ups and an SCD type 2 machine dimension.
- **BI:** Direct Lake semantic model with DAX measures, a coverage grid for completeness, six report pages stored as PBIR JSON and deployed from Git.
- **Market prices:** Dataflow Gen2 loads public day-ahead prices into a Warehouse; a T-SQL procedure builds a cost mart that reaches the model through a OneLake shortcut.
- **Real time:** the same stream into an Eventhouse; KQL functions (deduplication, state runs, overruns) drive a Real-Time Dashboard and Activator e-mails: one per heat-up overrun, one per failed pipeline run.
- **Operations:** daily pipeline with retries, table maintenance (OPTIMIZE, V-Order, VACUUM) and an explicit model reframe; paid F4 capacity scripted and paused when idle (about €12 so far).

DECISIONS WORTH DISCUSSING

- Why the gateway sends live data first after an outage, and how silver copes with late data.
- Why deduplicate on (meter, measurement time) rather than on the sequence number.
- Energy from counter deltas, and plausibility rules that turned 217,000 kWh of phantom energy into 10 flagged discontinuities.
- Why an SCD2 dimension needs its surrogate key in every relationship, and how that breaks row-level security if you get it wrong.
- Direct Lake framing: why the model is reframed once at the end of the pipeline.
- Transformations as a tested Python package, shipped to Fabric as a wheel in an Environment.
- Why an alert must fire on a transition, not a state: the first heat-up rule sent 91 e-mails.

TECHNOLOGY

Microsoft Fabric (Eventstream, Lakehouse, Spark notebooks, Environment, pipelines, Direct Lake, Power BI, Eventhouse and KQL, Real-Time Dashboard, Activator, Dataflow Gen2, Warehouse and T-SQL, OneLake shortcuts, Git integration, REST API) · PySpark, Delta Lake · Python, pytest · TypeScript, Node.js, node-opcua, vitest · Docker · GitHub · Azure CLI



Next: CI on GitHub Actions and a demo video. Details, screenshots and the 43 written design decisions are in the repository.